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Accuracy of generic musculoskeletal models in predicting the functional roles of muscles in human gait

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ABSTRACT

Biomechanical assessments of muscle function are often performed using a generic musculoskeletal model created from anatomical measurements obtained from cadavers. Understanding the validity of using generic models to study movement biomechanics is critical, especially when such models are applied to analyze the walking patterns of persons with impaired mobility. The aim of this study was to evaluate the accuracy of scaled-generic models in determining the moment arms and functional roles of the lower-limb muscles during gait. The functional role of a muscle was described by its potential to contribute to the acceleration of a joint or the acceleration of the whole-body center of mass. A muscle's potential acceleration was defined as the acceleration induced by a unit of muscle force. Dynamic simulations of walking were generated for four children with cerebral palsy and five age-matched controls. Each subject was represented by a scaled-generic model and a model developed from magnetic resonance (MR) imaging. Calculations obtained from the scaled-generic model of each subject were evaluated against those derived from the corresponding MR-based model. Substantial differences were found in the muscle moment arms computed using the two models. These differences propagated to calculations of muscle potential accelerations, but predictions of muscle function (i.e., the direction in which a muscle accelerated a joint or the center of mass and the magnitude of the muscle's potential acceleration relative to that of other muscles) were consistent between the two modeling techniques. Our findings suggest that scaled-generic models and image-based models yield similar assessments of muscle function in both normal and pathological gait.

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1. Introduction

Biomechanical assessments of muscle function are often performed using generic models of the body. Generic musculoskeletal models (e.g., Amis et al., 1979; Delp et al., 1990; Holzbaur et al., 2005; Klein Horsman et al., 2007; Ward et al., 2009) are typically based on measurements obtained from cadaveric specimens or medical images, and are presumed to be representative of able-bodied adults. Scaled-generic models are created by scaling body-segment anthropometry, joint geometry, and muscle-tendon attachment sites in the model to corresponding parameters in individual subjects. Such scaling requires measurements of body mass and limb lengths, the latter often estimated from surface-mounted markers or from direct measurements of distances between anatomical landmarks. The accuracy of scaled-generic models in representing musculoskeletal anatomy remains largely untested, mainly because of the time-intensive nature of creating benchmark models from subject-specific medical images. Scheys et al. (2008a, 2008b) assessed the mechanical advantage (i.e., moment arms) of individual muscles computed from scaled-generic and image-based models, and found significant errors due to inter-subject anatomical variability and bony abnormalities. However, no study to our knowledge has determined the effects of scaling on muscle function during any task. Understanding the validity of using scaled-generic models to study movement biomechanics is critical, especially for patients with conditions such as cerebral palsy (CP), where bone and muscle abnormalities are the rule rather than the exception.

Magnetic resonance (MR) scanners provide high-resolution images that allow visualization of anatomical structures *in vivo*. From these images, geometrical model parameters can be found to describe joint locations and orientations, muscle attachment sites, and bony torsion (Scheys et al., 2006; Blemker et al., 2007). MR-based models are able to reproduce bone geometry accurately (Smith et al., 1989) and have been shown to closely approximate muscle moment-arm measurements obtained from cadaver specimens (Murray et al., 1998; Arnold et al., 2000). Although processing algorithms can expedite extraction of bone structures (Hoad and Martel, 2002; Zoroofi et al., 2004) and muscle paths (Scheys et al., 2009),

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image-based models remain prohibitively expensive to implement in clinical environments due to MR scan costs and image processing time (Viceconti et al., 2006; Blemker et al., 2007).

Musculoskeletal computer models and simulations yield information that cannot be obtained non-invasively by direct measurement. Muscle forces can be determined using inverse-dynamics-based optimization, dynamic optimization, and EMG-driven modeling (Zajac, 1993; Lloyd and Besier, 2003; Erdemir et al., 2007; Pandy and Andriacchi, 2010); and neuromuscular coordination patterns can be revealed through muscle contributions to joint accelerations and the acceleration of the body's center of mass (Anderson and Pandy, 2003; Neptune et al., 2004; Liu et al., 2006; Pandy et al., 2010). Unfortunately, optimization models cannot readily be applied to study the gait patterns of individuals with neurological conditions such as CP, which are characterized by sub-optimal neural function. The EMG-driven modeling approach also has limitations, principal among these being difficulties associated with normalization of the EMG data and non-invasive measurement of deep-lying muscle activity.

Geometric assessments of muscle function are not affected by the aforementioned limitations. The functional role of a given muscle can be quantified by a potential contribution to the acceleration of a joint or the center of mass, defined as the acceleration that a unit muscle force (1 N) would generate (Schwartz and Lakin, 2003; Arnold et al., 2005; Hicks et al., 2007, 2008). This quantity provides insight into how the relative positions of muscles, bones, and joints influence a muscle's capacity to generate motion. Among the most important accelerations in walking are the vertical and fore-aft accelerations of the center of mass (representing support and forward progression, respectively), and the flexion–extension accelerations of the ipsilateral hip, knee, and ankle joints (Anderson and Pandy, 2003; Arnold et al., 2005; Liu et al., 2006).

The overall goal of this study was to evaluate the accuracy of scaled-generic musculoskeletal models relative to MR-based models in determining the potential contributions of the lower-limb muscles to joint accelerations and the acceleration of the body's center of mass during walking. This evaluation was performed for a cohort of children with CP and a cohort of age-matched controls. Our specific aims were firstly, to evaluate errors in muscle moment-arm calculations produced by scaled-generic models; and secondly, to explore whether these errors propagate to patterns of muscle function predicted by scaled-generic models.

2. Methods

Scaled-generic and MR-based musculoskeletal models were developed for nine study participants. Four subjects (2 males and 2 females; age, 10.5 ± 3.7 years; height, 133.2 ± 14.8 cm; weight, 30.6 ± 4.7 kg) diagnosed with spastic diplegic CP were identified from medical records held at the Royal Children's Hospital in Melbourne. The other five subjects (3 males and 2 females; age, 10.1 ± 2.0 years; height, 141.1 ± 12.6 cm; weight 40.6 ± 12.2 kg) were able-bodied controls. Subjects were aged between 7 and 15 years, were able to ambulate barefoot and unassisted, displayed a symmetrical gait pattern, and had not undergone extensive orthopedic surgery or selective dorsal rhizotomy. The CP patients were classified by an experienced physiotherapist as level II or III under the Gross Motor Function Classification System (Palisano et al., 1997). Further subject details are shown in Table 1. Project approval was obtained from the relevant Human Research Ethics Committees, and written informed consent was provided according to the committees' guidelines.

All subjects underwent a clinical gait analysis and a lower-body MR scan at the Royal Children's Hospital in Melbourne. Joint motion and ground reaction forces were recorded simultaneously as each subject walked at his/her preferred speed on level ground. Three-dimensional trajectories of retro-reflective skin-mounted markers were measured using a 10-camera motion capture system (VICON, Oxford Metrics Group, Oxford, U.K., 100 Hz). Ground reaction forces and moments were measured using six strain-gauged force plates (AMTI, Watertown, MA, USA, 1000 Hz). Marker trajectories were filtered using a Woltring (1986) filter (MSE 15).

For each subject, the pelvis and lower limbs (from the iliac crest to the distal talus) were imaged using either a 3.0 T or 1.5 T MR scanner (Siemens, Erlangen, Germany). Four series of three-dimensional PD Water-Suppressed Fast Spin Echo images were obtained, providing 0.9 mm contiguous transverse slices with an in-plane resolution of 0.9 mm $\times$ 0.9 mm and a field of view of 360 mm $\times$ 360 mm.

In both the scaled-generic and the MR-based models, the body was represented as a 10-segment, 23-degree-of-freedom linkage, actuated by 92 muscle-tendon units. The head, arms, and torso were lumped together and modeled as a single rigid body, which articulated with the pelvis via a ball-and-socket back joint. Each hip was modeled as a ball-and-socket joint, and each knee as a modified one-degree-of-freedom planar joint (Yamaguchi and Zajac, 1989). Each talo-crural joint, subtalar joint, and metatarso-phalangeal joint was modeled as a hinge. Each muscle was represented as one or more line segments acting between an origin point on the proximal segment and an insertion point on the distal segment. The paths of muscles that wrapped over underlying structures were modeled using via points (Delp et al., 1990).

Scaled-generic models were developed by scaling each segment of the generic model of Delp et al. (1990) to each subject's dimensions. Body mass and inter-marker distances computed from a standing trial (Table 2) were used as the scaling factors. The model parameters scaled were: bone geometry; segmental mass properties; locations of the joint centers and joint axes of rotation; muscle paths; muscle-fiber lengths; and tendon rest lengths.

MR-based models were developed based on the geometry of the pelvis and bones of the right and left lower limbs (i.e., femur, patella, tibia, fibula, and talus) as acquired from the MR images. Image segmentation of bone meshes was performed using Amira (Visage Imaging, Berlin, Germany). Coordinate systems for the pelvis, femur, and tibia segments and the joint centers/axes for the hip, knee, and talo-crural joints were defined using Rhinoceros (McNeel, Seattle, WA, USA). The hip joint was defined as the center of the femoral head; the knee axis was defined as the trans-epicondylar line; and the talo-crural axis was defined as the centerline of a cylinder wrapping over the proximal talus.

Femoral anteversion was incorporated into the MR-based models, and tibial torsion was incorporated into all models, using the methods described by Arnold et al. (2000) and Hicks et al. (2007), respectively. Inclusion of femoral anteversion modified the muscle attachments on the proximal femur, whereas inclusion of tibial torsion modified the orientation of the talo-crural axis and the muscle attachments on the distal tibia and fibula. Femoral anteversion was measured as the angle in the transverse plane between the femoral neck axis and the condylar axis of the distal femur (Ruby et al., 1979; Murphy et al., 1987). Tibial torsion was measured as the angle in the transverse plane between the knee and talo-crural joint flexion–extension axes (Staheli, 1993).

Table 1
Details of the participants used in this study.

Subject	Gender	Age (years)	BMI (kg/m^2)	GMFCS level	Femoral anteversion (L/R) (deg)	External tibial torsion (L/R) (deg)	Prior surgery
CP1	F	8.0	18.4	III	33/25	18/24	Bilateral psoas tenotomy, bilateral percutaneous adductor tenotomy
CP2	M	15.1	15.6	II	32/28	30/3	Nil
CP3	M	11.8	14.5	II	26/31	35/25	Nil
CP4	F	6.9	21.2	II	27/30	12/20	Nil
NC1	M	12.1	25.3	N/A	21/23	32/28	Nil
NC2	F	8.8	17.9	N/A	30/33	20/28	Nil
NC3	F	7.5	16.1	N/A	32/40	28/25	Nil
NC4	M	12.1	18.4	N/A	26/23	20/27	Nil
NC5	M	10.1	22.4	N/A	26/28	28/32	Nil

Femoral anteversion and external tibial torsion were measured during clinical examination. The symbols CP and NC denote cerebral palsy patients and control subjects, respectively.

Muscle-tendon paths in the MR-based models were defined from anatomical landmarks identified for the following muscles: gluteus maximus, gluteus medius, iliopsoas, hamstrings, rectus femoris, vasti, gastrocnemius, and soleus. These muscles are a combination of the major muscles of the lower limb and represent those that typically function abnormally in CP and require medical treatment. All other muscle paths were identical with those in the corresponding scaled-generic model. Muscle wrapping points were modified from the corresponding scaled-generic model so that each muscle wrapped smoothly over the underlying bones.

OpenSim software (Delp et al., 2007) was used to generate and analyze simulations of walking derived for both models using a single representative gait trial of each subject. Inverse Kinematics and Residual Reduction algorithms were used in conjunction with the measured marker trajectories and ground reaction forces and moments to obtain a set of dynamically consistent joint angles. For each subject, these joint angles were calculated using the MR-based model. The results were then applied to both the MR-based model and the corresponding scaled-generic model, so that comparisons could be attributed to the modeling technique and not to differences in joint motion. Muscle moment arms about the hip, knee, and ankle were computed using the tendon excursion method (An et al., 1984).

A perturbation analysis (Liu et al., 2006) was used to determine the potential (per-unit-force) contributions of individual muscles to the angular accelerations of the hip, knee, and ankle, and to the vertical and fore-aft accelerations of the body's center of mass. This analysis, performed for single-leg stance, required a muscle-actuated simulation, with actuator excitations calculated by applying the Computed Muscle Control method (Thelen et al., 2003). We note here that the potential contributions to acceleration are independent of muscle excitations and muscle forces, and are determined only by the configuration of the skeleton and the muscle lines-of-action during the time period of interest. At every 0.01 s of the gait simulation, the force of each muscle was increased by 1 N and the simulation then integrated forward for 0.03 s (Liu et al., 2006). Ground reaction forces were allowed to change by the use of spring-damper elements positioned at the center-of-pressure of the foot, with linear elements active throughout the stance phase and torsional elements activated only when the foot was flat on the ground. For muscles represented by more than one muscle path, the potential contributions to acceleration were computed using a weighted average based on the maximum isometric force of each muscle path included in the generic model. Potential contributions to acceleration were normalized using the method described in Appendix A.

Differences in moment arm between each scaled-generic model and its corresponding MR-based model were based on the mean values calculated for single-leg stance and were assessed using Bland–Altman 95% limits of agreement (Bland and Altman, 1986). Differences in potential muscle contributions to acceleration between the two models were quantified using a root-mean-square (RMS) difference, which was expressed as a percentage of the peak value obtained from the MR-based model.

Table 2
Marker pairs used to scale body segments.

Segment	Coordinate	Inter-marker distances used for segment scaling
Torso	X,Y,Z	Average of RPSI-MAN and LPSI-MAN
Pelvis	X	Average of RASI-RPSI and LASI-LPSI
Pelvis	Y,Z	RASI-LASI
R/L thigh	X,Y,Z	ASI-KNE (R/L)
R/L shank	X,Y	KNE-ANK (R/L)
R/L shank	Z	KNE-MED (R/L)
R/L foot	X,Y,Z	HEE-TOE (R/L)

X, Y, and Z represent the segment dimensions in the anterior–posterior, superior–inferior, and medial–lateral directions, respectively. Symbols used to represent retro-reflective marker locations are PSI (posterior superior iliac spine), ASI (anterior superior iliac spine), MAN (manubrium), KNE (lateral femoral condyle), ANK (lateral malleolus), MED (medial malleolus), HEE (posterior calcaneous) and TOE (2nd metatarsal).

3. Results

The average walking speeds of the control subjects and CP patients were 1.3 ± 0.2 m/s and 1.0 ± 0.1 m/s, respectively. The simulation results displayed consistency with the experimental data and with the results of previous studies. Simulated joint angles of the pelvis, hips, knees, and ankles were within 4° of the experimental data for all subjects. Sagittal-plane joint angles displayed by the CP patients differed from the mean data obtained for the control subjects by up to 20° (Fig. 1). The potential muscle contributions to the hip- and knee-joint accelerations obtained from the MR-based models of the control subjects were consistent with results reported by Hicks et al. (2008) (Fig. 2).

There were substantial differences in the moment arms computed for several muscles using the two modeling techniques (Table 3). Most noticeably, scaled-generic models overestimated the knee moment arms of the hamstring muscles by up to 31.2%

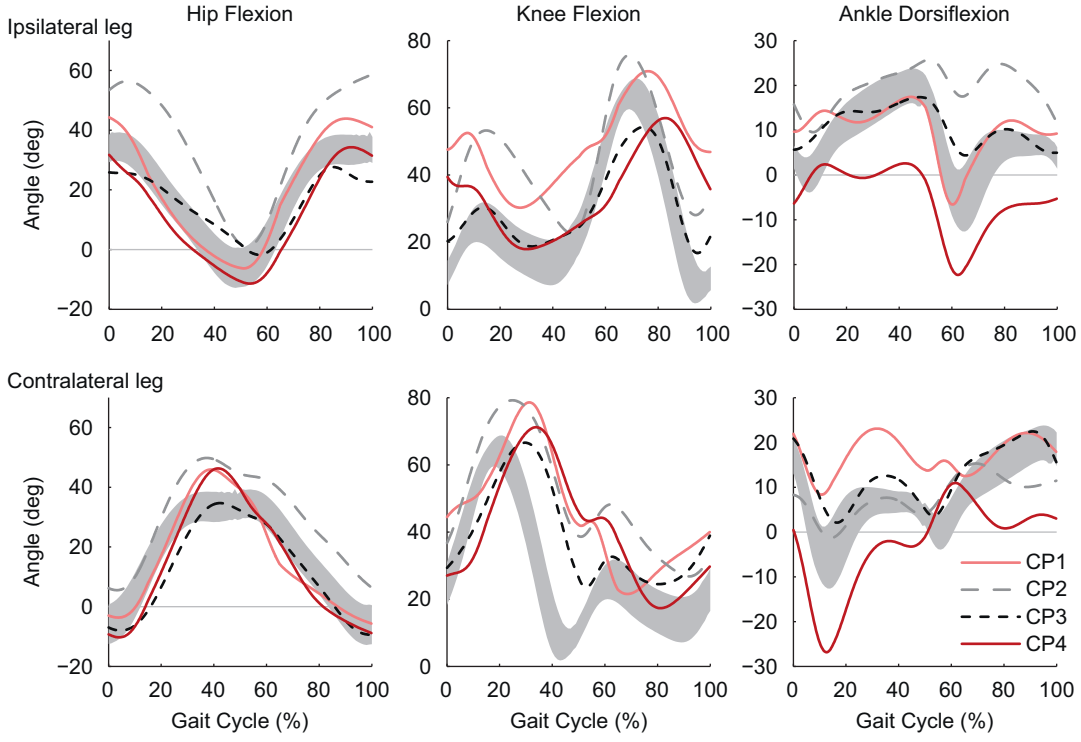


Fig. 1. Sagittal-plane joint angles recorded for four CP patients walking at an average speed of 1.0 m/s. The gray area represents one standard deviation above and below the mean recorded for five control subjects walking at an average speed of 1.3 m/s. 0% and 100% represent heel-strike of the ipsilateral leg.

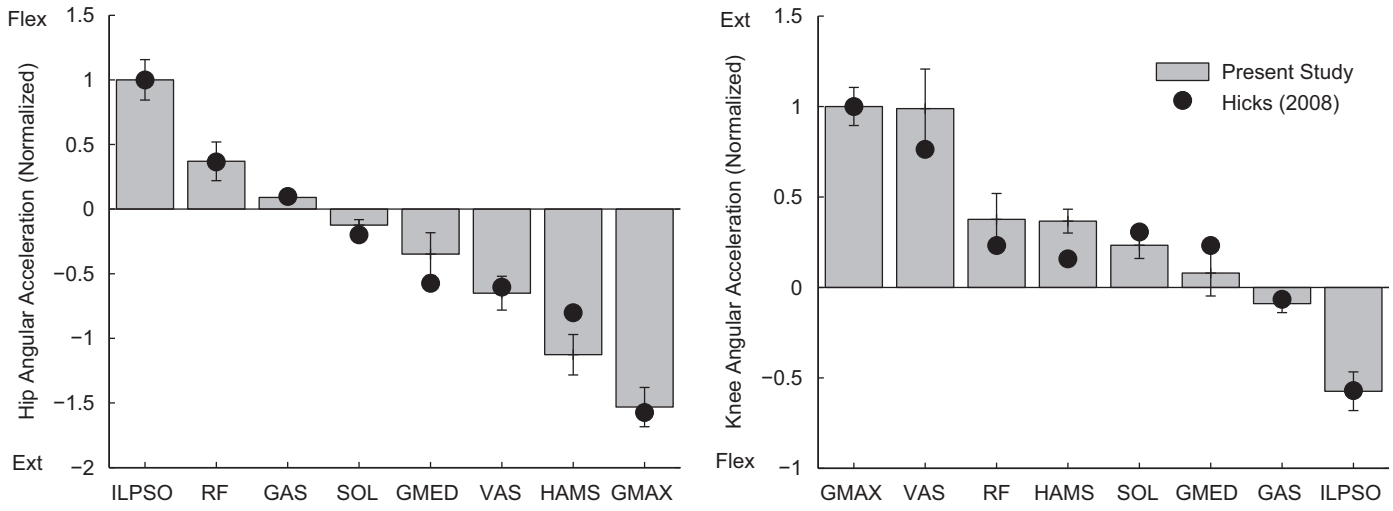


Fig. 2. Mean muscle potential contributions to hip- and knee-joint angular accelerations for single-leg stance averaged across five control subjects. Results were obtained using MR-based models and normalized to the largest positive potential contribution. A muscle's potential contribution to joint acceleration was defined as the angular acceleration that the muscle generates per unit of force. Normalization between subjects of varying anthropometry was performed as described in Appendix A. Results reported by Hicks et al. (2008) are based on the average gait pattern measured for 83 able-bodied children and were obtained using a generic musculoskeletal model and a different analysis technique from the one adopted in the present study. Symbols used to represent the various muscle groups are ILPSO (iliacus and psoas major combined), RF (rectus femoris), GAS (medial and lateral gastrocnemius combined), SOL (soleus), GMED (gluteus medius), VAS (vastus lateralis, vastus intermedius, and vastus medialis combined), HAMS (semimembranosus, semitendinosus, and biceps femoris long head combined), and GMAX (gluteus maximus).

and the hip moment arm of rectus femoris by up to 25.1%. There was high variability in moment arm differences for both cohorts.

Results obtained from the two modeling techniques were consistent in the predictions of the relative capacities of individual muscles to accelerate the lower-limb joints and the center-of-mass during single-leg stance. In particular, the two models produced consistent results with respect to the direction in which a given muscle accelerated a joint or the center of mass (Figs. 3 and 4, bar charts); the magnitude of that muscle's potential acceleration relative to that of other muscles (Figs. 3 and 4); and the timing and shape of the potential acceleration induced by that muscle (Figs. 3 and 4, line graphs). Quantitative differences between the mean potential accelerations computed by the two models were evident, however, for both cohorts. The mean RMS differences in the potential muscle contributions to the hip, knee, and center-of-mass accelerations obtained from the two models ranged from 5.7% to 37.5% for the control subjects and from 5.0% to 28.8% for the CP patients (Table 4). The CP patients also displayed large RMS differences in muscle potential contributions to ankle-joint acceleration.

4. Discussion

To our knowledge, this is the first study to examine the accuracy of scaled-generic models in assessing the functional roles of muscles during gait. Our aim was to evaluate the accuracy of scaled-generic models in computing both muscle moment arms and the potential contributions of individual muscles to joint and center-of-mass accelerations in normal and pathological gait. The accuracy of these models was quantified by comparing their predictions against those obtained from MR-based models, each of which required around 10 additional hours of processing time, with the majority of this time insensitive to computing power. We found that for children with a range of anatomical abnormalities and mild gait deviations, scaled-generic musculoskeletal models do not accurately reproduce muscle moment arms, yet they yield muscle potential accelerations that are qualitatively similar to those obtained from MR-based models.

Several limitations must be considered when interpreting our results. First, only nine subjects were assessed due to the time-intensive nature of creating subject-specific models from MR images. Our results were consistent among subjects of varying anthropometry and degrees of bony torsion; however, further research is needed before our findings can be applied more broadly. Second, all subjects with CP displayed relatively mild alterations in their gait patterns. It is possible that more severe gait abnormalities, particularly those involving extreme amounts of hip and knee flexion, would not be assessed with the same degree of precision by scaled-generic models. Third, for each subject, joint angles were calculated from the MR-based model. Inaccurate joint parameters in scaled-generic models can lead to errors in calculated joint angles and therefore net joint moments (Reinbolt et al., 2007), but multi-phase optimization methods for parameter calibration (e.g., Reinbolt et al., 2008) can be applied to minimize these errors. Fourth, several muscles were represented by multiple lines of action and several broad attachment sites by a single point. For example, single points representing the broad origins of gluteus medius and soleus are difficult to define objectively, which may have affected our comparisons between modeling techniques for these muscles. Fifth, our analysis was confined to single-leg stance because, as noted by Hicks et al. (2008), induced accelerations computed during double support exhibit large discontinuities, which are a result of the perturbation method applied, as well as heel-strike transient effects. Finally, the RMS differences for both cohorts showed substantial variability for many of the muscle potential accelerations of the joints and the center of mass (Table 4, standard deviations), and so many of the values presented for the 'mean RMS difference' may not apply to larger cohorts. Despite this high variability, muscle function was qualitatively similar between the two modeling techniques for the gait simulations that displayed the least agreement.

We found substantial errors in the muscle moment arms computed using scaled-generic models, with inter-subject averages as high as 31.2% (Table 3). This result is consistent with the findings of Scheys et al. (2008a, 2008b) who reported hip flexion–extension moment arm errors as high as 36.3%, and it also confirms the variability of inter-individual musculoskeletal geometry reported

Table 3
Differences in the mean moment arms of selected muscles crossing the hip, knee, and ankle obtained from scaled-generic and MR-based models for the single-leg stance phase of gait.

		Control subjects			Cerebral palsy patients		
		Average difference in mean moment arm (± standard deviation)	Mean moment arm (MR models) (mm)	95% limits of agreement (Bland–Altman) (mm)	Average difference in mean moment arm (± standard deviation)	Mean moment arm (MR models) (mm)	95% limits of agreement (Bland–Altman) (mm)
Gluteus maximus anterior	Hip extension	0.4% (± 19.1%)	−34.2	−12.2:13.0	−9.2% (± 12.4%)	−31.6	−5.2:10.2
Gluteus maximus middle	Hip extension	−7.6% (± 14.1%)	−48.2	−9.0:17.0	−2.1% (± 4.6%)	−39.5	−2.5:3.8
Gluteus maximus posterior	Hip extension	−11.5% (± 9.2%)	−71.6	−4.5:21.4	−2.4% (± 11.9%)	−59.2	−11.8:15.9
Semimembranosus	Hip extension	1.8% (± 11.5%)	−48.1	−12.8:11.7	4.2% (± 12.5%)	−41.5	−9.5:7.6
Semitendinosus	Hip extension	13.7% (± 13.4%)	−50.6	−20.6:7.1	11.1% (± 10.4%)	−45.6	−12:4.7
Biceps femoris (long head)	Hip extension	9.3% (± 10.3%)	−51.7	−16.7:7.2	7.8% (± 10.2%)	−46.7	−10.5:5.4
Iliacus	Hip flexion	3.5% (± 12.6%)	34.3	−7.1:8.6	7.0% (± 11.1%)	28.9	−3.9:6.7
Psoas major	Hip flexion	1.8% (± 11.0%)	33.5	−6.5:6.9	12.7% (± 20.9%)	27.1	−5.1:9.2
Rectus femoris	Hip flexion	25.1% (± 15.5%)	34.5	1.0:15.0	9.4% (± 17.5%)	34.5	−7.7:11.5
Semimembranosus	Knee flexion	13.5% (± 7.5%)	−29.5	−7.4: −0.4	12.5% (± 6.2%)	−28.2	−7.6:1.9
Semitendinosus	Knee flexion	11.1% (± 12.6%)	−35.4	−11.1:4.5	10.0% (± 6.1%)	−34.1	−6.6:1.5
Biceps femoris (long head)	Knee flexion	17.4% (± 5.5%)	−25.6	−6.9: −1.9	31.2% (± 34.3%)	−22.5	−13.4:4.1
Medial gastrocnemius	Knee flexion	2.6% (± 11.6%)	−16.9	−3.3:2.9	−10.5% (± 19.8%)	−17.5	−4.7:8.7
Lateral gastrocnemius	Knee flexion	−2.9% (± 12.4%)	−18.0	−4.2:6.0	−3.6% (± 14.8%)	−16.1	−4.1:5.6
Rectus femoris	Knee extension	1.7% (± 6.2%)	42.2	−5.2:6.0	1.4% (± 7.2%)	37.7	−4.5:5.3
Vastus medialis	Knee extension	−1.3% (± 7.5%)	38.6	−7.7:5.9	−1.2% (± 7.3%)	34.0	−4.8:4.2
Vastus intermedius	Knee extension	−0.7% (± 6.8%)	40.2	−6.9:5.6	−0.5% (± 7.0%)	35.8	−4.8:4.5
Vastus lateralis	Knee extension	−2.3% (± 7.9%)	39.2	−8.6:5.8	−1.9% (± 7.1%)	35.1	−5.1:4.0
Medial gastrocnemius	Ankle PF	4.6% (± 2.2%)	−35.9	−3.3:0.0	2.2% (± 2.8%)	−32.8	−2.4:1.2
Lateral gastrocnemius	Ankle PF	5.5% (± 1.3%)	−35.5	−3.3: −0.6	3.5% (± 4.1%)	−31.8	−3.3:1.5
Soleus	Ankle PF	3.9% (± 1.3%)	−34.3	−2.4: −0.3	2.4% (± 3.9%)	−30.8	−2.9:1.7

For each cohort (control subjects and cerebral palsy patients), results were averaged for both limbs. Results for each muscle are expressed as a percentage of the mean value obtained from the MR-based model, which is also shown. The Bland–Altman 95% limits of agreement were calculated from the difference that resulted when scaled-generic model values were subtracted from MR-based model values. The symbol PF denotes plantarflexion.

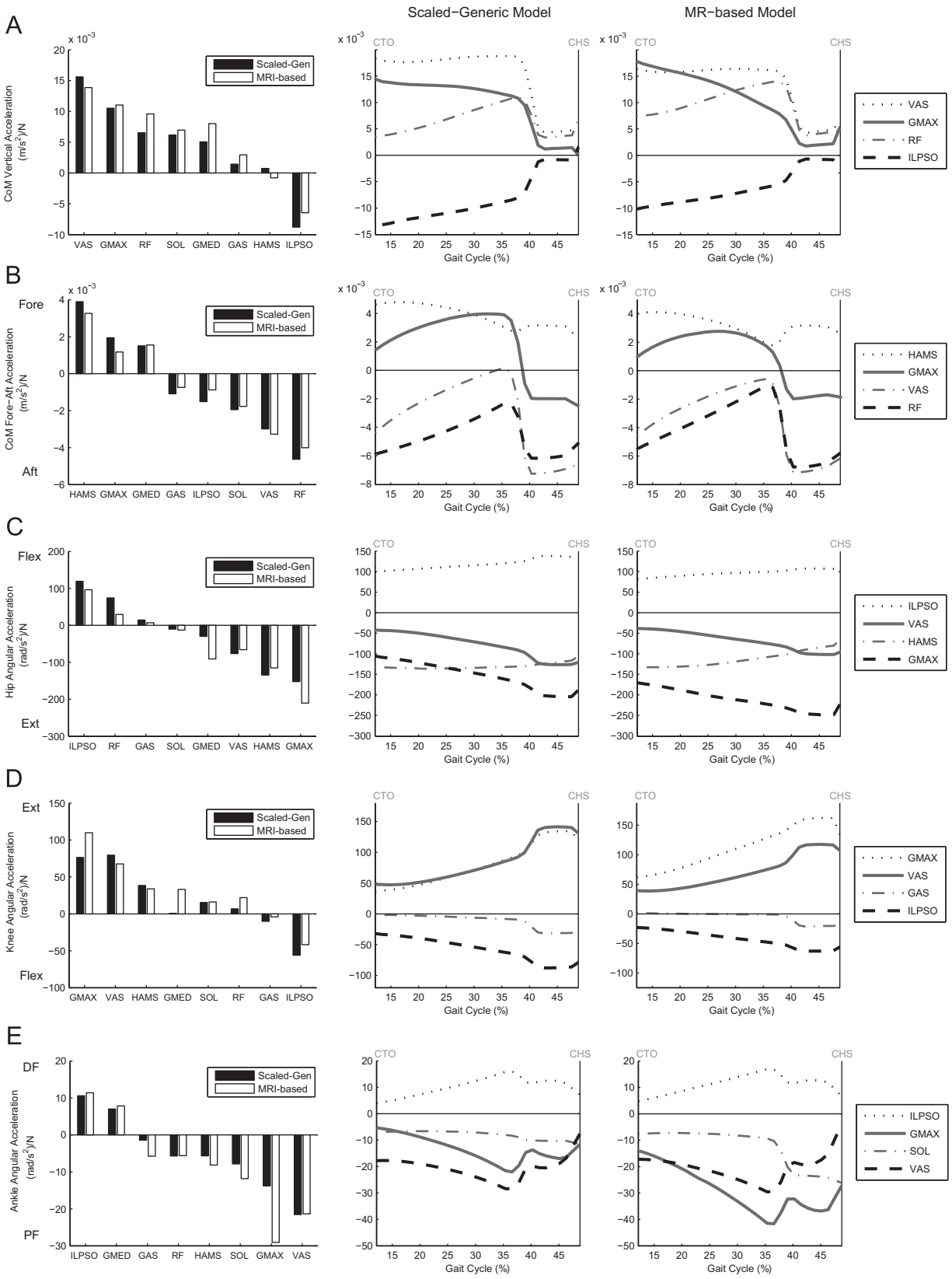


Fig. 3. Potential muscle contributions to joint and center-of-mass accelerations calculated for the control subjects. Each horizontal plot shows the results for the subject that displayed the largest differences between modeling methods for a given joint or center-of-mass acceleration. The data from subject NC3 showed the largest differences for all joint and center-of-mass accelerations, and so all data shown in the figure apply to subject NC3. The plots describe the potential muscle contributions to (A) vertical acceleration of the body's center of mass, (B) fore-aft acceleration of the body's center of mass, (C) hip flexion–extension acceleration, (D) knee flexion–extension acceleration, and (E) ankle plantarflexion–dorsiflexion acceleration. The bar charts show the mean potential accelerations computed for single-leg stance for the eight muscles investigated. The line graphs show the potential accelerations from contralateral toe-off (CTO) to contralateral heel-strike (CHS) for the muscles that displayed large positive and negative mean values.

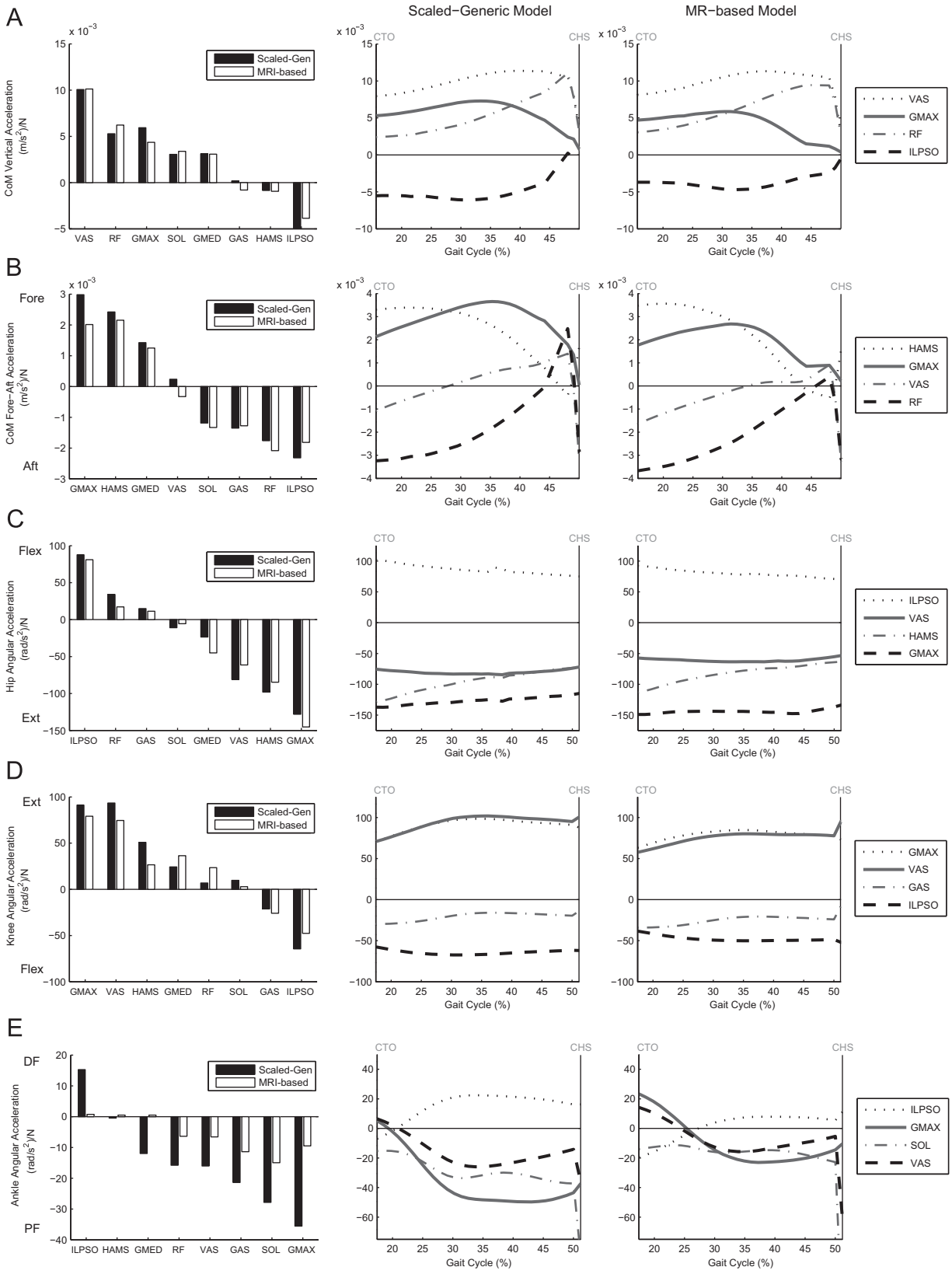


Fig. 4. Potential muscle contributions to joint and center-of-mass accelerations calculated for the CP patients. Each horizontal plot shows the results for the subject that displayed the largest differences between modeling methods for a given joint or center-of-mass acceleration. The data presented in plots A and B are from subject CP2, while the data presented in plots C, D, and E are from subject CP4. The plots describe the potential muscle contributions to (A) vertical acceleration of the body's center of mass, (B) fore-aft acceleration of the body's center of mass, (C) hip flexion–extension acceleration, (D) knee flexion–extension acceleration, and (E) ankle plantarflexion–dorsiflexion acceleration. The bar charts show the mean potential accelerations computed for single-leg stance for the eight muscles investigated. The line graphs show the potential accelerations from contralateral toe-off (CTO) to contralateral heel-strike (CHS) for the muscles that displayed large positive and negative mean values.

Table 4
RMS differences in potential muscle contributions to joint accelerations and to the vertical and fore-aft accelerations of the body's center of mass (CoM) determined from scaled-generic and MR-based models.

Scaled-generic model versus MR-based model: RMS percentage difference in potential accelerations					
	Hip flexion/extension	**Knee flexion/extension**	**Ankle DF/PF**	**Vertical acc. of CoM**	**Fore-aft acc. of CoM**
Control subjects					
Gluteus maximus	13.7% (±10.6%)	9.5% (±6.4%)	21.6% (±13.8%)	20.0% (±13.0%)	32.9% (±13.4%)
Gluteus medius	22.1% (±13.7%)	37.5% (±10.7%)	24.6%	14.1% (±9.8%)	13.6% (±5.5%)
Iliopsoas	11.6% (±6.4%)	14.3% (±10.1%)	10.6%	13.0% (±7.3%)	31.6% (±15.9%)
Hamstrings	13.8% (±6.1%)	18.3% (±5.6%)	8.6% (±3.6%)	22.9% (±12.1%)	12.5% (±7.3%)
Rectus femoris	26.2% (±16.3%)	21.1% (±7.3%)	15.5% (±14.0%)	20.3% (±12.2%)	11.4% (±5.9%)
Vasti	9.4% (±5.7%)	6.9% (±4.1%)	11.4% (±5.8%)	6.3% (±2.6%)	8.7% (±4.7%)
Gastrocnemius	14.3% (±10.3%)	23.6% (±11.2%)	12.5% (±10.1%)	14.6% (±11.8%)	10.6% (±3.2%)
Soleus	N/A	5.7% (±3.9%)	12.9% (±10.2%)	6.6% (±5.8%)	8.6% (±5.2%)
Average	**15.9%**	**17.1%**	**14.7%**	**14.7%**	**16.2%**
Cerebral palsy patients					
Gluteus maximus	8.1% (±4.1%)	6.7% (±3.8%)	69.7% (±37.7%)	16.3% (±9.3%)	23.5% (±14.4%)
Gluteus medius	20.7% (±16.8%)	23.5% (±11.1%)	19.5% (±2.5%)	5.0% (±1.1%)	7.3% (±2.5%)
Iliopsoas	13.7% (±10.2%)	23.4% (±12.1%)	69.5%	9.0% (±12.7%)	15.6% (±12.7%)
Hamstrings	13.6% (±8.7%)	28.8% (±23.5%)	N/A	27.3% (±19.0%)	9.0% (±4.1%)
Rectus femoris	25.3% (±10.1%)	23.8% (±9.4%)	N/A	12.6% (±8.3%)	11.8% (±5.7%)
Vasti	13.8% (±10.3%)	10.2% (±9.9%)	18.2% (±16.1%)	6.1% (±2.9%)	9.3% (±10.5%)
Gastrocnemius	10.6% (±3.2%)	9.5% (±6.0%)	39.8% (±19.8%)	16.8% (±21.3%)	13.0% (±8.3%)
Soleus	N/A	N/A	36.0%	6.1% (±4.3%)	10.4% (±7.6%)
Average	**15.1%**	**18.0%**	**42.1%**	**12.4%**	**12.5%**

Results for each muscle are expressed as a percentage of the peak value obtained from the MR-based model. The differences shown are the average (and standard deviation) for each cohort of subjects obtained for the single-leg stance phase of gait. Results from multiple lines of action were combined for the following muscles: gluteus maximus (anterior, middle, and posterior compartments), gluteus medius (anterior, middle and posterior compartments), iliopsoas (iliacus and psoas major), hamstrings (semimembranosus, semitendinosus, and biceps femoris long head), vasti (vastus lateralis, vastus intermedius and vastus medialis) and gastrocnemius (medial and lateral compartments). The symbols DF and PF represent dorsiflexion and plantarflexion, respectively. Muscles were included in the table only if the subject's MR-based model generated a peak value greater than 0.00125 $(m/s^2)/N$ for potential accelerations of the center of mass or greater than 20 $(rad/s^2)/N$ for potential joint angular accelerations. Standard deviations are not shown if only one subject's data fit the above criteria.

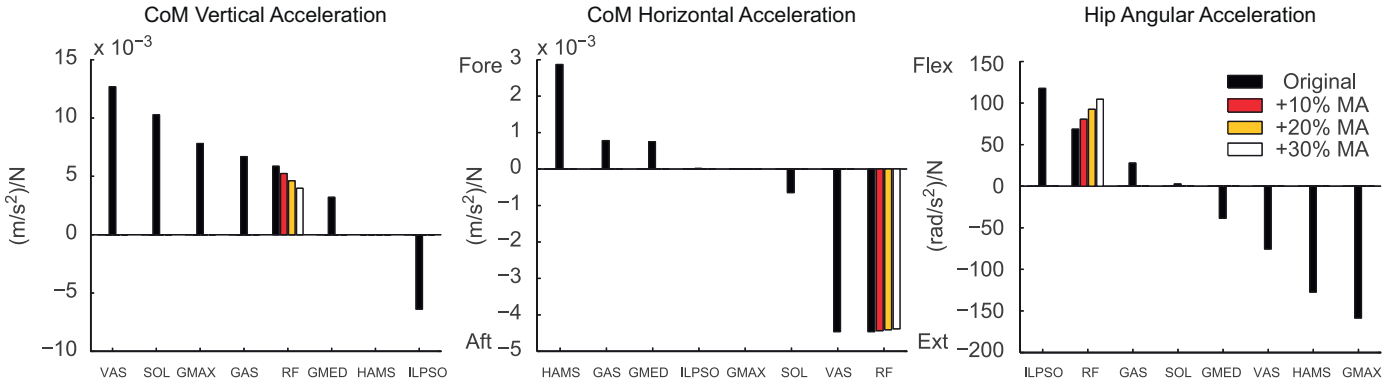


Fig. 5. Effect of increasing the hip-flexion moment arm (MA) of rectus femoris on the potential contributions of this muscle to the vertical acceleration of the body's center of mass (CoM); the fore-aft acceleration of the body's center of mass; and the flexion–extension acceleration of the hip. Results were obtained using an MR-based model of one of the control subjects. The moment arm was increased by as much as 30% relative to the nominal value. The moment arm of rectus femoris at the hip was altered by moving the muscle's origin point anteriorly, which allowed uniform moment arm increases over the range of motion while maintaining a similar muscle length.

by others (Brand et al., 1982; Duda et al., 1996). These errors propagated to calculations of muscle potential accelerations, which describe a muscle's capacity to accelerate the body.

Although RMS errors of up to 37.5% were obtained for the potential accelerations of the hip, knee, and center of mass predicted by the scaled-generic models (Table 4), the functional roles of the muscles were qualitatively similar to that predicted by the MR-based models, even for those gait simulations that displayed the least agreement. This does not mean that accurate calculations of muscle moment arms are not important for analyses of movement; indeed, they are crucial in the calculation of muscle forces (Herzog, 1992; Raikova and Prilutsky, 2001). The results of the present study suggest that substantial changes in a muscle's moment arm alone do not alter predictions of the functional role of that muscle with respect to (1) the direction in which the muscle accelerates a joint or the center of mass; (2) the magnitude of the muscle's potential acceleration relative to that of other muscles; and (3) the timing and shape of the potential acceleration induced by the muscle (Figs. 3 and 4).

To further illustrate this result, we perturbed the moment arm of a representative muscle in the model (i.e., rectus femoris) and studied the effect of this change on the muscle's potential to accelerate the lower-limb joints and center of mass (Fig. 5). Changes of up to 30% in the hip moment arm of rectus femoris produced changes of similar magnitudes in the calculated values of the muscle potential accelerations, but the functional role of the muscle was unchanged with respect to the directions in which rectus femoris accelerated the hip and the center of mass;

the magnitude of the acceleration induced by rectus femoris relative to that induced by the other muscles; and the timing and shape of rectus femoris' potential accelerations of the hip and center of mass (not shown in Fig. 5).

Increased errors in muscle potential accelerations were observed at the ankle joint (Table 4), which may be due to the relatively simple foot–ground contact model used and the proximity of the ankle to the foot–ground contact surface. More detailed models of foot–ground contact are needed to obtain improved estimates of muscle-induced accelerations at the ankle (Dorn et al., 2011).

Although a muscle's potential contribution to acceleration is geometrical in nature, this measure may be coupled with calculations of muscle force to predict dynamic function. Further research is needed to develop a robust and clinically feasible methodology to estimate muscle forces in pathological populations. Given their demonstrated accuracy and low cost, scaled-generic musculoskeletal models may assist clinicians in the diagnosis and treatment of individuals with gait abnormalities.

Conflict of interest statement

The authors do not have any financial or personal relationships with other people or organizations that could inappropriately influence their work.

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Appendix A. Supplementary Materials

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jbiomech.2011.05.023.

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